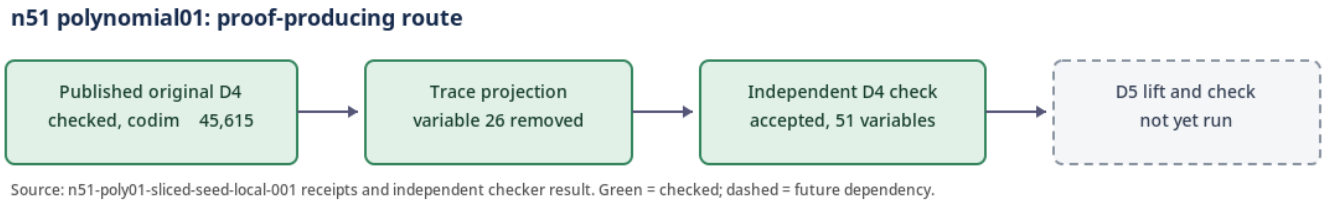
A checked 51-variable D4 seed for the n51 degree-five search



Source: n51-poly01-sliced-seed-local-001 receipts and independent checker result. Green = checked; dashed = future dependency.

The trace slice of **fresh-n51-poly-01** now has an independently accepted degree-4 nonmembership map. The checker rebuilt 51 equations, verified the trace substitution b0 = a0 at variable 26, and accepted C0-C3. It reported codimension at least **45,615** and low-kernel dimension 10,194.

Stage	Outcome	Frozen limit
Projection	Exit 0; 1,553,961,056 bytes	4 GiB; 1,800 s
Independent D4 check	Accepted; codim at least 45,615	16 GiB; 3,600 s

The projected certificate has SHA-256 `8c0936e5cad0b84b769ab77f6a5a0d499f36f1356153463a50db2b618c8971e4`. It derives from the published original D4 certificate with raw SHA-256 `c160792a7530b20dd17affcdabafb8d95494eed51c8ee6d818af3212f3e304f6`. The exact launch, source, original plan, and independent descent hashes are pinned in the companion evidence.json and committed protocol.

The projection receipt writer encountered a Python 3.9 API mismatch after the projection completed. Its receipt was recovered from the immutable output and transcript; the producer was not rerun. The checker completed once with empty stderr. The additional reservation was USD 3 on the shared existing Spot host.

This is a checked D4 seed for a D5 lift. The earlier original n51 D4 certificate had already established LFD at least 5 on this named instance. A D5 producer and independent D5 check are separate future steps; no LFD at least 6 claim follows from this report.

Full evidence, file paths, digests, and replay commands: report.md, evidence.json, and the raw/ directory alongside this PDF.